

DirtBike Maintenance System

IoT Predictive Maintenance for Dirt Bikes

CSC 494-001

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1. Problem & Importance

Problem

Dirt bikes face dust, high RPMs, crashes → neglected maintenance causes 80–90% of engine "blow-ups"

- Low oil → seizure, scored bearings
- Low coolant → overheating, warped heads
- Clogged air filter → power loss, abrasive wear

Why important?

Prevents costly rebuilds, mid-ride failures, enhances safety on

2. Project Topic

Software Focus

Build IoT application on ESP32 hardware:

- Real-time sensor data processing & calibration
- Mobile-friendly dashboard with gauges, trends & alerts

Hardware Used

- Core: ESP32 microcontroller (WiFi + Bluetooth)
- Sensors: Waterproof ultrasonic (oil/coolant levels)
- Differential pressure sensor (air-box restriction)
- Power: DC-DC converter from 12V bike battery

Communication

- WiFi: Cloud sync for historical data when in range
- Bluetooth: Connection to smartphone app for dashboards, gauges, and alerts

3. Two Topics to Learn with AI

Software

Build IoT app on ESP32 hardware:

- Build dashboard
- Alert logic + buffering

Hardware

ESP32 sensor interfacing & calibration

- Ultrasonic & differential pressure

4. Time Usage – Two Iterations

Iteration 1 (Sprint 1)

Learn → build

- Sensor reading/calibration
- Basic dashboard + thresholds

Iteration 2 (Sprint 2)

Full features

- Alerts (push/email)
- Trends, local buffering
- Vibration/dust testing

5. Final Goal (End of Semester)

- **Working prototype** mounted & demonstrated
- **Research paper**: show findings, bugs, fixes, and w
- **Portfolio piece**: GitHub repo, code, schematics, demo video
- **Future extension** potential: vibration sensor

Questions?